

Database Management Systems: Assignment-7

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Q1. Consider the following table. Mention the expected super key, candidate key, primary key, and alternate key. Give syntax for the same.

Patients Table

Patient ID	Patient Name	Sex	Age	MedInsuranceNo	AadharNo
01	Sheela	F	23	Med0291	839292319012
02	Rehan	M	21	Med8421	123456789012
03	Anay	M	56	Med4203	848298469201
04	Mahira	F	42	Med4792	724723021922
05	Nishant	M	12	Med8419	472038311099

Answer:

- **Super Keys:** {PatientID}, {MedInsuranceNo}, {AadhaarNo}, {PatientID, PatientName}, ...
- **Candidate Keys:** {PatientID}, {MedInsuranceNo}, {AadhaarNo}
- **Primary Key:** {PatientID}
- **Alternate Keys:** {MedInsuranceNo}, {AadhaarNo}

Query:

```
CREATE TABLE Patients (PatientID INT PRIMARY KEY, PatientName VARCHAR(50), Sex CHAR(1), Age INT, MedInsuranceNo VARCHAR(20) UNIQUE, AadhaarNo VARCHAR(20) UNIQUE);
```

Q2. Create an Emp table with the following fields:

(EmpNo, EmpName, Job, Basic, DA, HRA, PF, GrossPay, NetPay)

(Calculate DA as 30% of Basic and HRA as 40% of Basic)

- a) Mention the expected super key, candidate key, primary key, and alternate key. Give syntax for the same.

Answer:

- **Super Keys:** {EmpNo}, {EmpNo, EmpName}, {EmpNo, Job}, ...
- **Candidate Keys:** {EmpNo}
- **Primary Key:** {EmpNo}
- **Alternate Keys:** None

Query:

```
CREATE TABLE Emp (EmpNo INT PRIMARY KEY, EmpName VARCHAR(50), Job
VARCHAR(30), Basic DECIMAL(10,2), DA DECIMAL(10,2), HRA DECIMAL(10,2), PF
DECIMAL(10,2), GrossPay DECIMAL(10,2), NetPay DECIMAL(10,2));
```

- b) Insert Five Records and calculate GrossPay and NetPay.

Query:

```
INSERT INTO Emp (EmpNo, EmpName, Job, Basic, DA, HRA, PF, GrossPay, NetPay)
VALUES
(1, 'Amit', 'Manager', 20000, 0.3*20000, 0.4*20000, 0.1*20000, NULL, NULL),
(2, 'Sneha', 'Clerk', 8000, 0.3*8000, 0.4*8000, 0.1*8000, NULL, NULL),
(3, 'Rohit', 'Analyst', 12000, 0.3*12000, 0.4*12000, 0.1*12000, NULL, NULL),
(4, 'Meena', 'Clerk', 9000, 0.3*9000, 0.4*9000, 0.1*9000, NULL, NULL),
(5, 'Karan', 'Manager', 25000, 0.3*25000, 0.4*25000, 0.1*25000, NULL, NULL);
```

```
UPDATE Emp SET GrossPay = Basic + DA + HRA, NetPay = Basic + DA + HRA - PF;
```

Output:

EmpNo	EmpName	Job	Basic	DA	HRA	PF	GrossPay	NetPay
1	Amit	Manager	20000.00	6000.00	8000.00	2000.00	34000.00	32000.00
2	Sneha	Clerk	8000.00	2400.00	3200.00	800.00	13600.00	12800.00
3	Rohit	Analyst	12000.00	3600.00	4800.00	1200.00	20400.00	19200.00
4	Meena	Clerk	9000.00	2700.00	3600.00	900.00	15300.00	14400.00
5	Karan	Manager	25000.00	7500.00	10000.00	2500.00	42500.00	40000.00

- c) Display the employees who's Basic is lowest in each department.

Query:

```
SELECT e.EmpNo, e.EmpName, e.Job, e.Basic FROM Emp e WHERE Basic = (SELECT
MIN(Basic) FROM Emp WHERE Job = e.Job);
```

Output:

EmpNo	EmpName	Job	Basic
1	Amit	Manager	20000.00
2	Sneha	Clerk	8000.00
3	Rohit	Analyst	12000.00

- d) If NetPay is less than <Rs. 10,000 add Rs. 1200 as special allowances. Display the employees whose GrossPay lies between 10,000 & 20,000.

Query:

```
UPDATE Emp SET NetPay=NetPay+1200 WHERE NetPay<10000;
```

Output:

EmpNo	EmpName	Job	Basic	DA	HRA	PF	GrossPay	NetPay
1	Amit	Manager	20000.00	6000.00	8000.00	2000.00	34000.00	32000.00
2	Sneha	Clerk	8000.00	2400.00	3200.00	800.00	13600.00	12800.00
3	Rohit	Analyst	12000.00	3600.00	4800.00	1200.00	20400.00	19200.00
4	Meena	Clerk	9000.00	2700.00	3600.00	900.00	15300.00	14400.00
5	Karan	Manager	25000.00	7500.00	10000.00	2500.00	42500.00	40000.00

Query:

```
SELECT * FROM Emp WHERE GrossPay BETWEEN 10000 AND 20000;
```

Output:

EmpNo	EmpName	Job	Basic	DA	HRA	PF	GrossPay	NetPay
2	Sneha	Clerk	8000.00	2400.00	3200.00	800.00	13600.00	12800.00
4	Meena	Clerk	9000.00	2700.00	3600.00	900.00	15300.00	14400.00

- e) Display all the employees who earn maximum salary.

Query:

```
SELECT * FROM Emp WHERE NetPay = (SELECT MAX(NetPay) FROM Emp);
```

Output:

EmpNo	EmpName	Job	Basic	DA	HRA	PF	GrossPay	NetPay
5	Karan	Manager	25000.00	7500.00	10000.00	2500.00	42500.00	40000.00

Q3. Consider the insurance database given below. The primary keys are made bold and the data types are specified.

PERSON(driver_id:string , name:string , address:string)

CAR(regno:string , model:string , year:int)

ACCIDENT(report_number:int , accd_date:date ,location:string)

OWNS(driver_id:string , regno:string)

ARTICIPATED(driver_id:string,regno:string,report_number:int,damage_amount:int)

- a) Mention the expected super key, candidate key, primary key, unique key and alternate key. Give syntax for the same.

Answer:

PERSON:

- **Super Keys:** {driver_id}, {driver_id, name}, ...
- **Candidate Keys:** {driver_id}
- **Primary Key:** {driver_id}
- **Unique Key:** {driver_id}
- **Alternate Keys:** None
- **Query:** CREATE TABLE PERSON (driver_id VARCHAR(20) PRIMARY KEY, name VARCHAR(50), address VARCHAR(100));

CAR:

- **Super Keys:** {regno}, {regno, model}, ...
- **Candidate Keys:** {regno}
- **Primary Key:** {regno}
- **Unique Key:** {regno}
- **Alternate Keys:** None
- **Query:** CREATE TABLE CAR (regno VARCHAR(20) PRIMARY KEY, model VARCHAR(50), year INT);

ACCIDENT:

- **Super Keys:** {report_number}, {report_number, accd_date}, ...
- **Candidate Keys:** {report_number}
- **Primary Key:** {report_number}
- **Unique Key:** {report_number}
- **Alternate Keys:** None
- **Query:** CREATE TABLE ACCIDENT (report_number INT PRIMARY KEY, accd_date DATE, location VARCHAR(100));

OWNS:

- **Super Keys:** {driver_id, regno}
- **Candidate Keys:** {driver_id, regno}
- **Primary Key:** {driver_id, regno}
- **Unique Key:** {driver_id, regno}
- **Alternate Keys:** None
- **Query:** CREATE TABLE OWNS (driver_id VARCHAR(20), regno VARCHAR(20), PRIMARY KEY(driver_id, regno), FOREIGN KEY(driver_id) REFERENCES PERSON(driver_id), FOREIGN KEY(regno) REFERENCES CAR(regno));

ARTICIPATED:

- **Super Keys:** {driver_id, regno, report_number}, {driver_id, regno, report_number, damage_amount}
- **Candidate Keys:** {driver_id, regno, report_number}
- **Primary Key:** {driver_id, regno, report_number}
- **Unique Key:** {driver_id, regno, report_number}
- **Alternate Keys:** None
- **Query:** CREATE TABLE ARTICIPATED (driver_id VARCHAR(20), regno VARCHAR(20), report_number INT, damage_amount INT, PRIMARY KEY(driver_id, regno, report_number), FOREIGN KEY(driver_id, regno) REFERENCES OWNS(driver_id, regno), FOREIGN KEY(report_number) REFERENCES ACCIDENT(report_number));

- b) Enter at least five tuples for each relation based on a given query.

Query:

```
INSERT INTO PERSON VALUES
('D001','Amit','Mumbai'),
('D002','Sneha','Delhi'),
('D003','Rohit','Pune'),
('D004','Meena','Chennai'),
('D005','Karan','Kolkata');
```

Output:

driver_id	name	address
D001	Amit	Mumbai
D002	Sneha	Delhi
D003	Rohit	Pune
D004	Meena	Chennai
D005	Karan	Kolkata

Query:

```
INSERT INTO CAR VALUES
('MH01AB1234','Swift',2010),
('DL05XY5678','Innova',2008),
('PN12CD3456','Swift',2008),
('TN07EF7890','Fortuner',2009),
('WB09GH1234','Swift',2011);
```

Output:

regno	model	year
DL05XY5678	Innova	2008
MH01AB1234	Swift	2010
PN12CD3456	Swift	2008
TN07EF7890	Fortuner	2009
WB09GH1234	Swift	2011

Query:

```
INSERT INTO ACCIDENT VALUES
(11,'2008-03-15','Mumbai'),
(12,'2008-07-20','Delhi'),
(13,'2009-05-10','Pune'),
(14,'2008-12-01','Chennai'),
(15,'2010-01-25','Kolkata');
```

Output:

report_number	accd_date	location
11	2008-03-15	Mumbai
12	2008-07-20	Delhi
13	2009-05-10	Pune
14	2008-12-01	Chennai
15	2010-01-25	Kolkata

Query:

```
INSERT INTO OWNS VALUES
('D001','MH01AB1234'),
('D002','DL05XY5678'),
('D003','PN12CD3456'),
('D004','TN07EF7890'),
('D005','WB09GH1234');
```

Output:

driver_id	regno
D001	MH01AB1234
D002	DL05XY5678
D003	PN12CD3456
D004	TN07EF7890
D005	WB09GH1234

Query:

```
INSERT INTO ARTICIPATED VALUES
('D001','MH01AB1234',11,15000),
('D002','DL05XY5678',12,20000),
('D003','PN12CD3456',13,18000),
('D004','TN07EF7890',14,22000),
('D005','WB09GH1234',15,25000);
```

Output:

driver_id	regno	report_number	damage_amount
D001	MH01AB1234	11	15000
D002	DL05XY5678	12	20000
D003	PN12CD3456	13	18000
D004	TN07EF7890	14	22000
D005	WB09GH1234	15	25000

- c) Demonstrate how you, Update the damage amount for the car with specific regno in the accident with report number 12 to 25000. Find the total number of people who owned cars that were involved in accidents in the year 2008.

Query:

```
UPDATE ARTICIPATED SET damage_amount=25000 WHERE regno='DL05XY5678'
AND report_number=12;
```

Output:

driver_id	regno	report_number	damage_amount
D001	MH01AB1234	11	15000
D002	DL05XY5678	12	25000
D003	PN12CD3456	13	18000
D004	TN07EF7890	14	22000
D005	WB09GH1234	15	25000

Query:

```
SELECT COUNT(driver_id) AS total_people FROM ARTICIPATED a join ACCIDENT ac
ON a.report_number = ac.report_number WHERE YEAR(ac.accd_date)=2008;
```

Output:

total_people
3

- d) Find the number of accidents in which cars belonging to a specific model were involved.

Query:

```
SELECT COUNT(a.report_number) AS accident_count FROM ARTICIPATED a join CAR  
c ON a.regno = c.regno WHERE c.model='Swift';
```

Output:

accident_count
3

- e) Create Sales table with the following fields:(Sales No, Salesname, Branch, Salesamount, DOB)

1. Insert five records.

Query:

```
INSERT INTO Sales VALUES  
(1,'Amit','Mumbai',50000,'1985-12-21'),  
(2,'Sneha','Delhi',60000,'1987-11-15'),  
(3,'Rohit','Mumbai',55000,'1990-12-05'),  
(4,'Meena','Delhi',70000,'1988-12-10'),  
(5,'Karan','Kolkata',65000,'1992-07-25');
```

Output:

SalesNo	SalesName	Branch	SalesAmount	DOB
1	Amit	Mumbai	50000.00	1985-12-21
2	Sneha	Delhi	60000.00	1987-11-15
3	Rohit	Mumbai	55000.00	1990-12-05
4	Meena	Delhi	70000.00	1988-12-10
5	Karan	Kolkata	65000.00	1992-07-25

2. Calculate total salesamount in each branch.

Query:

```
SELECT Branch, SUM(SalesAmount) AS TotalSales FROM Sales GROUP BY Branch;
```

Output:

Branch	TotalSales
Delhi	130000.00
Kolkata	65000.00
Mumbai	105000.00

3. Calculate average salesamount in each branch.

Query:

```
SELECT Branch, AVG(SalesAmount) AS AvgSales FROM Sales GROUP BY Branch;
```

Output:

Branch	AvgSales
Delhi	65000.000000
Kolkata	65000.000000
Mumbai	52500.000000

4. Display all the salesmen, DOB who are born in the month of December as day in character format i.e. 21-Dec-09

Query:

```
SELECT SalesName, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB FROM Sales WHERE MONTH(DOB) = 12;
```

Output:

SalesName	DOB
Amit	21-Dec-85
Rohit	05-Dec-90
Meena	10-Dec-88

5. Display the name and DOB of salesman in alphabetical order of the month.

Query:

```
SELECT SalesName, DOB FROM Sales ORDER BY MONTH(DOB);
```

Output:

SalesName	DOB
Karan	1992-07-25
Sneha	1987-11-15
Amit	1985-12-21
Rohit	1990-12-05
Meena	1988-12-10